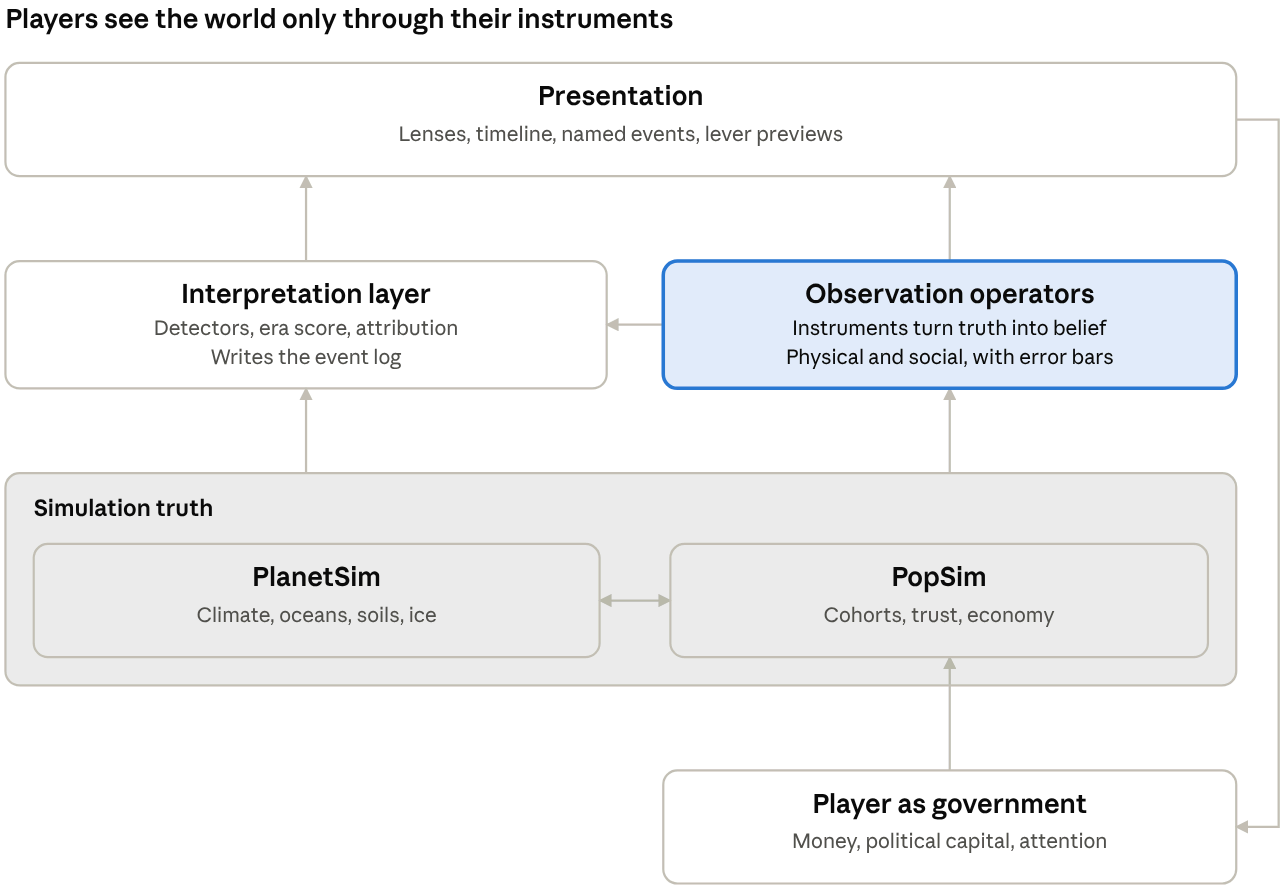
Planetary Civilization Simulator — Systems Architecture

Oct 4, 2026 · @frederick gianni

Overview

Civ VI writes outcomes as rules; we simulate causes. So every borrowed Civ mechanic becomes a reading or a naming of simulated state, never a rule of its own. Five boxes carry this: two simulation cores, one interpretation layer, one presentation layer, and the player acting as government.



system architecture · 5 boxes, 2 observation channels

Detectors read truth so the world really changes; everything the player sees and decides on passes through the observation operators first.

Design rules

1. The interpretation layer only reads simulation state; it never writes physics back.
2. Player actions reach the world only through PopSim and the economy.
3. Every named event links to a cause in the simulation, so the debrief gets causal chains for free.
4. The simulation exports truth; the player sees only what their instruments measure, on the planet and in society alike.

Civ VI borrowings, re-implemented

Civ VI mechanic	How Civ does it	How we do it	Layer
Disasters with an upside	Dice roll per tile	Floods from rain and river routing; fertility from sediment in the soil model	PlanetSim + interpretation
Per-civ CO_2	Counted per building	Emissions tagged at source; linear attribution ledger	Interpretation
Climate phases	Fixed 7-step script	Thresholds on a continuous field; real tipping nonlinearities	Interpretation
Lowland warnings	Precomputed tiles	Forked low-resolution ensemble with uncertainty bands	Interpretation + presentation
Lenses	Static overlays	Rendered fields, sharpness set by observation coverage	Presentation
Eurekas	Hardcoded triggers	Research priority follows exposure; learning-by-doing	PopSim + levers
Loyalty	City pressure from neighbours	Cohort trust; exit, voice or loyalty	PopSim
Golden and Dark Ages	Era score from scripted moments	Two-axis score from simulated change over an era	Interpretation

Civ VI mechanic	How Civ does it	How we do it	Layer
World Congress	Leader agendas	Other civs are PopSim instances; votes follow their opinion	PopSim + levers

What we avoid from Civ: climate that is easy to ignore, free-riding without consequence, victory conditions blind to the planet, and tile-stacking puzzles.

PlanetSim: field exposure

PlanetSim publishes immutable, typed snapshots through a fixed contract; nothing reads its internals. Truth and observation are separate channels.

Snapshots

- At the end of each tick: tick id, simulation time, RNG state, field buffers.
- Double-buffered or copy-on-write: readers process tick N while the simulation computes N+1. The simulation never waits.
- Readers get zero-copy `std::span` views over float32 structure-of-arrays.

Field registry

```
struct FieldDesc {
  FieldId id; Unit unit;            // SI only
  GridId grid; Cadence cadence;     // fast / slow loop
  Stat stat;                        // instant, mean, accumulated
  Range valid; uint32_t version;
};
```

- Readers request fields by id; the schema is versioned.
- Tests inject synthetic fields through the same interface, so detectors run without physics. Parallel agents can work against the fixed contract.

Accumulators, not samples

- Rain, emissions, runoff and energy fluxes are totals since the last snapshot. Nothing is lost when the simulation substeps.
- Rolling windows, climatologies and percentiles belong to the interpretation layer.

Grid, regions, reductions

- Fields stay on the native mesh grid.
- Precomputed region masks: land and ocean, river basins, civ territory, lowland bands, PopSim cohort regions.
- A reduction API (area-weighted mean, sum, max, percentile over a mask) covers most detector needs without full-field scans.

Truth versus observed

- The simulation exports truth only.
- An observation operator produces observed fields with a variance field: Observed = H(truth, network) + noise. It uses station and satellite coverage, sensor error and science level.
- Detectors fire on truth and flag whether the event was observed. Lenses, attribution estimates and projections use the observed channel.
- Knowledge gating lives in one place: the operator.

Tendency diagnostics

- Key fields (temperature, soil moisture, river discharge, sea level) also export budget terms: each process's share of the change.
- These give event records their cause weights, for example a flood split between rainfall anomaly, upstream deforestation runoff and sea level.
- Optional per field, behind a flag.

Path to Godot

- GDExtension packs lens fields into downsampled textures on the render thread.
- Timeline keyframes are compressed low-resolution copies every N years.
- Godot never touches full-resolution buffers.

Determinism

- Snapshot + seed + player-input log reproduce any run bit-for-bit.
- Golden-snapshot regression tests catch silent physics changes from agents editing the core in parallel.

PopSim: trust and loyalty

PopSim models a few hundred cohorts per civ, each with several kinds of trust, and turns dissatisfaction into behaviour through Hirschman's Exit, Voice, Loyalty.

Cohorts

- Defined by region × class or sector × age band × urban or rural.
- Sector matters most for climate: coal miners and coastal city workers react to policy very differently.
- The debrief can sample a few representative individuals from a cohort for stories.

State per cohort

Variable	Meaning	Drives
Competence trust	Does the government deliver?	Votes, compliance
Integrity trust	Does it do what it said?	Response to promises
Fairness trust	Are burdens shared justly?	Protest, compliance
Science trust	Belief in climate science	Acceptance of climate framing and policy
Social cohesion	Trust between cohorts	Polarisation, cooperation
Loyalty	Attachment to civ vs region	Secession pressure, migration
Expectations	Adaptive reference point	How outcomes are judged

Update rules

- **Performance vs expectations:** income, health and safety are compared with the reference point. Losses weigh about twice as much as gains. Expectations drift toward current conditions.
- **Promise ledger:** each media statement stores target, deadline and salience. Broken promises cost integrity in proportion to salience; kept ones earn less.
- **Trust asymmetry:** trust falls several times faster than it rises (Slovic).
- **Procedural fairness:** a cohort's policy burden is compared with others' and with its own responsibility from the attribution ledger (Tyler). An efficient carbon tax that spares frequent flyers still triggers a gilets-jaunes-type reaction.
- **Disasters:** a short rally effect, then the response is judged. Voters reward relief more than preparedness (Healy and Malhotra), a built-in dilemma.
- **Social influence:** opinions spread between connected cohorts with a bounded-confidence model. Polarisation emerges; media reach and slant shape the network.

Outputs

- **Voice:** protests, strikes, votes. Scales with dissatisfaction × belief that speaking up works.
- **Exit:** internal or external migration, capital flight, secession pressure. Secession needs low central loyalty, strong regional identity and an attractive neighbour.
- **Loyalty:** policy compliance, tax compliance, tolerance of sacrifice.

Key coupling with the implementation gap:

```
\text{policy effectiveness} = \text{design} \times \text{compliance}
(\text{trust}, \text{fairness})
```

Elections

- Vote shares come from trust and outcome judgments.
- A lost election installs a new government with a honeymoon bonus. Integrity damage stays attached to institutions, so the civilization keeps paying.

Calibration sources: European Social Survey, World Values Survey, Eurobarometer, OECD Trust Survey; Hirschman (1970), Levi and Stoker (2000), bounded-confidence opinion dynamics.

Population knowledge: fog on society

The government never sees PopSim's truth. It sees a belief state built from social instruments, exactly as it sees the planet through stations and satellites. The difference: social instruments are biased, not just noisy, and measuring a society changes it.

Social observation operator

```
\text{Belief}_{c,v} = H_s(\text{truth}, \text{instruments},
\text{institutions}) + \text{noise} + \text{bias}
```

- Stored per cohort c and variable v: estimate, variance, bias, staleness.
- Same interface as the physical operator: an `ObservedField` with a variance field. Social instruments are new sensor implementations behind it.
- The belief state works like a Kalman filter. Predict: the government's own model of society advances the estimate and uncertainty grows. Update: each new measurement narrows it.

- The government's predictive model is itself imperfect. Social-science research improves it and reduces systematic bias.

Instruments

Instrument	Measures	Cadence	Weakness	Available from
Census	Cohort sizes, location, occupation	10 years	Stale; undercounts informal and remote cohorts	1880
Prefect and governor reports	Regional mood, unrest	Months	Tell the centre what it wants to hear	1880
Press monitoring	Issue salience, elite opinion	Weeks	Urban and owner bias	1880
Petitions and elections	Grievances; vote intention revealed	Event / term	Rare; elections reveal everything at once	1880
Statistics office	Prices, wages, employment, inequality	Quarter / year	Only as honest as its independence	~1900, matures ~1950
Tax and admin records	Income, firm activity	Continuous	Formal economy only	Needs a tax system
Opinion polls	Trust, vote intention, salience	Weeks	Sampling error, non-response, social desirability	~1935
Focus groups	Why a cohort feels as it does	On demand	Small sample	~1950
Panel surveys	Trust dynamics over time	Years	Expensive	~1970
Digital sentiment analytics	Real-time mood	Hours	Loud, young, online bias; bots; rival manipulation	~2005

Instrument	Measures	Cadence	Weakness	Available from
Surveillance	Organising, unrest threats	Continuous	Destroys trust and integrity when exposed	Any era, authoritarian path

Mechanics

- **Invisible cohorts:** cohorts with no instrument coverage show as unknown. Unrest there arrives as a surprise.
- **Staleness:** uncertainty grows with time since the last measurement, so a census-only government drifts blind between counts.
- **Principal–agent bias:** officials' reports skew toward good news. Independent inspectors and an independent statistics office reduce the bias.
- **Reflexivity:** published polls shift opinion through bandwagon effects. Surveillance lowers trust. Massaged statistics diverge from lived experience; when the gap is exposed, integrity crashes.

Hooks into the other boxes

- **Interpretation:** social detectors fire on truth and carry an "anticipated?" flag. Riots you did not see coming become events; the debrief shows belief versus truth.
- **Presentation:** society lenses (trust, wealth, health, unrest risk) blur by coverage, like physical lenses. A timeline track compares poll forecasts with election results.
- **Levers:** budget for statistical capacity, an independent statistics office (an institution), the social-science research branch, and surveillance as a costly shortcut. Lever previews show winners and losers per cohort with error bars; uncovered cohorts show as unknown.

Interpretation layer

A read-only, deterministic module that turns each snapshot of PlanetSim and PopSim into events and metrics, written to an append-only event log. It runs on its own thread against snapshot N.

Threshold detectors

Every detector runs the same pipeline: signal → filter → threshold with hysteresis → persistence check → event record.

- **Signal:** a scalar from fields: global temperature anomaly, Arctic ice extent, discharge at a cell, crop yield, cohort trust.
- **Filter:** climate phases use a 20-year rolling mean so natural variability does not trigger them. Weather events use a percentile against local climatology.
- **Hysteresis:** on at one threshold, off at a lower one, so nothing flickers.
- **Persistence:** the condition must hold for N ticks.

Kind	Scope	Fires	Example
Phase	Global	Once	Climate phase 3 reached
Extreme event	Local	Per occurrence, by return period	"A 1-in-100 flood under the old climate, now 1-in-12"
Tipping	Global or regional	Once, plus early warnings	Ice sheet destabilising
Social	Cohort or region	Per occurrence	Riot, strike wave, secession movement

- A short pre-industrial control run at game start builds the baseline climatology for return periods.
- Tipping detectors watch the simulation's own nonlinear variables (ice–albedo, permafrost carbon). Early-warning signals (rising variance and autocorrelation) can be a science unlock.

Event record: type, time, region, magnitude, return period, weighted causes from tendency diagnostics, an observed flag (physical) or anticipated flag (social).

Era score

- **Window:** about one generation (25 years). The score measures change over the era, not level, so late recovery counts.
- **Human axis:** wellbeing, inequality, trust, promises kept.
- **Planet axis:** ecosystems, soils and oceans, change in committed warming, resilience (adaptation coverage relative to exposure).
- **Future generations:** a weight on committed impacts makes today's choices show up in a later era's score.

	Planet up	Planet down
Human up	Golden	Gilded (the tempting trap)
Human down	Stewardship (often costs an election)	Dark

- **Effects:** feeds PopSim optimism and trust baseline; unlocks era-specific policy options. Never changes physics.

Attribution ledger

- **Tagged at source:** emissions by civ × region × sector × gas × year. Aggregates only, so it stays small.
- **Linear impulse response:** each civ's emission history is convolved with the carbon-cycle response, then mapped to forcing and warming. Linearity makes contributions sum exactly to the total, as in real TCRE-based attribution.
- **Two views:** production-based and consumption-based (following trade). The gap is a diplomatic weapon.
- **Damage linkage:** extreme events link to warming through the fraction of attributable risk, then split by each civ's warming share. Example output: "Civ X caused 18% of the added flood risk in your delta." This feeds loss-and-damage talks.

```
\text{FAR} = 1 - \frac{P_0}{P_1}
```

P_0 is the event's probability in the baseline climate; P_1 its probability now.

- **Truth versus estimate:** the true ledger always exists. Players see an estimate whose error bars shrink with science and observation coverage. Rival civs can publish disputed figures.

Presentation layer

The presentation layer reads only observed field textures and the event log. It never queries a simulation, which keeps it swappable like a mod.

Lenses

- **Definition:** field source + colour mapping + knowledge gate + uncertainty style. Lens definitions are separate from style packs, so a cartoon mod restyles without changing meaning.
- **Rendering:** fields arrive as textures on the planet mesh; a Godot shader blends the overlay with the base art. A scan-sweep transition makes switching feel like turning on an instrument.
- **Coverage is visual:** unmeasured = fog or hatching; sparse = blur growing with distance from the nearest station; full = sharp. Stations and satellites show as dots.
- **Comparison slider:** then versus now, or with versus without you (using the pre-industrial control run).
- **Colour maps:** perceptually uniform, colour-blind-safe, diverging for anomalies.

Family	Lenses	Unlock (Industrial Dawn)
Physical	Temperature anomaly, rainfall, winds and currents as streamlines, sea ice, lowlands	Thermometers from 1880; satellites late
Living	Biodiversity, soils, ocean acidity	Ecology research
Human	Trust, wealth, health and air quality, exposure, unrest risk	Social instruments (see Population knowledge)
Responsibility	Emission plumes per civ	CO_2 record (Keeling-curve equivalent)
Future	Committed-future ghost coastline	Late science unlock

Timeline

- **Tracks:** era bands (Golden, Gilded, Stewardship, Dark), climate phase steps, event markers, elections and policies, promises kept or broken, poll forecasts versus election results, warming-stripes background.
- **Zoom:** from one turn to a century, matching the fast and slow feedback loops.
- **Past:** low-resolution keyframes every N years; scrubbing replays the planet. The debrief reuses this.
- **Future:** once unlocked, a dashed uncertainty cone past "now".

- **Causal threads:** selecting an event lights its causal links back through time, for example coal policy 1923 → emissions → warming share → flood 1971.

Event naming

- **Grammar:** [severity epithet] + [phenomenon] + [toponym] + [year], e.g. "The Great Delta Flood of 1931". Tiers come from magnitude and return period; recurrences take ordinals.
- **Scarcity:** only events above a significance threshold get a name, capped per era. The rest are summarised, e.g. "14 minor floods".
- **Normalisation as a lesson:** as frequency rises, names lose drama: "The Great Flood" → "the floods of '74" → "the flood season".
- **Framing follows knowledge and mood:** "act of nature" before attribution science, "climate-driven" with a share after it. Low-trust cohorts name events after the government that failed them. The media lever can contest framing.
- **Notification tiers:** toast for minor events; card with cause links for named events; cinematic fly-to with before/after for tipping points and surprise social events.
- **Localisation:** templates carry grammar tags (gender, elision), so "La Grande Crue du Delta de 1931" works from day one.

Population event naming

Social events need their own grammar because they have actors, grievances and outcomes, and because the people involved fight over the name. A natural event has one name; a social event has several, and which one wins is part of the game.

Grammar: [epithet] + [actor] + [action] + [grievance or symbol] + [place] + [year]. Any slot except action may be empty.

Type	Exit / Voice / Loyalty	Template	Example
Petition	Voice	[actor] petition on [grievance]	The Fishers' Petition on Harbour Rights
Demonstration	Voice	[place] march for [grievance]	The Delta March for Dikes
Strike	Voice	[actor] strike of [year]	The Miners' Strike of 1912

Type	Exit / Voice / Loyalty	Template	Example
Riot	Voice	[grievance] riots of [place]	The Bread Riots of Port-Sel
Movement	Voice	[symbol] movement	The Lantern movement
Uprising / secession crisis	Voice + exit	[place] [revolt / question]	The Northern Question
Migration wave	Exit	[epithet] [place] exodus	The Quiet Lowland Exodus
Capital or talent flight	Exit	The [decade] [drain / flight]	The Brain Drain of the 1990s
Election upset	Voice	The [place / actor] surprise of [year]	The Rural Surprise of 1936
Scandal	Voice	The [object] affair	The Census Affair
Collective effort	Loyalty	The [epithet] [season / effort]	The Solidarity Winter

Rules

- **Escalation ladder:** grumbling (never named) → petition → demonstration → strike → riot → uprising → secession crisis. An event that climbs the ladder is renamed at each step.
- **Epithets come from the simulation:** "Great" for size, "Long" for duration, "Bloody" only if casualties occurred, "Quiet" for exit-based events nobody protested.
- **Symbols come from cohort traits:** a movement adopts an emblem from its sector (miners' lanterns, fishers' nets, dockers' hooks), so names are generated, not authored.
- **Grievances come from cause weights:** the event record's causes fill the grievance slot, giving "Fuel-Tax Revolt", "Heat Strikes" or "Drought Exodus".
- **Life cycle:** emerging → escalating → peak → resolved (won, lost, repressed or absorbed). A provisional name runs while it is live; a settled name arrives at resolution. Winners tend to name it: "the Bread Riots" if repressed, "the Bread Revolution" if successful.
- **Contested names:** each event carries a government name, a movement name and, once settled, a historians' name. The media lever picks the official name. Each cohort adopts a name according to its trust, and the adoption share is itself an indicator of

where opinion sits. The debrief shows which name won.

- **Knowledge gating:** without instrument coverage of the cohort, the first name is generic, such as "the Northern disturbances". Grievance and actor slots fill only when reports, focus groups or polls reveal them.
- **Climate framing follows science:** before attribution science, a displacement is "the Delta Exodus" or an economic migration. After it, it can become "the Delta climate exodus", if the media lever allows.
- **Normalisation:** as with natural events, frequency drains drama: "the Great Dock Strike" → "the strikes of '74" → "strike season".
- **Decade names:** sustained patterns aggregate into decade epithets from PopSim aggregates, such as "the Restless Twenties" or "the Quiet Fifties".
- **Localisation:** the same tagged templates as natural events: "La Grève des Mineurs de 1912", "L'Exode silencieux des Basses-Terres".

Mod boundary

Lens definitions, palettes, icons, name templates and notification styles live in data packs. A style mod changes only these packs, never meaning.

Player levers as government

The player spends three scarce resources: money, political capital and attention. Every lever passes through politics and implementation before it touches PopSim or the economy.

Lever families

Family	Examples	Main cost	Main risk
Speech (media)	Promises, framing events, agenda-setting, information campaigns, media control	Political capital	Broken promises; censorship exposed by lived experience

Family	Examples	Main cost	Main risk
Budget	Income, consumption and carbon taxes; health, infrastructure, adaptation, relief, R&D spending; debt; carbon dividend	Money	Interest costs, default, unfair burdens
Regulation	Standards, bans, zoning (forest protection, managed retreat), building codes	Political capital	Low compliance
Investment	Power plants, grids, rail, dikes, cities	Money, time	Lock-in, stranded assets over 40-year lifetimes
Research portfolio	Energy, agriculture, climate science, social science, adaptation, observation networks	Money	Uncertain breakthroughs
Knowledge	Statistical capacity, independent statistics office, surveillance	Money; trust for surveillance	Blind spots; trust collapse if surveillance is exposed
Diplomacy	Treaties, trade, border carbon adjustments, aid, loss and damage, sanctions, World Congress votes	Political capital, money	Free-riders, retaliation
Institutions	Independent climate council, decentralisation, future-generations ombudsman	Rare era unlocks	Less flexibility later

Constraints

- **Political capital:** from trust, majority size and the post-election honeymoon. Contentious levers spend it.
- **Attention:** a limited number of major actions per turn, set by state capacity. Investing in administration raises it.

- **Veto players:** parties, unions, industry lobbies, green movements, regions. Each is backed by PopSim cohorts and can delay, water down or block. Coalition bargaining gets past them.
- **Implementation pipeline:** announced → enacted → implemented → complied with. Each stage has a delay and leakage, including corruption.
- **Timescale mismatch:** elections every 4–5 years, effects over decades. This is the game's central tension; the UI shows when each action's benefits become visible.

Lever previews

Before committing, the player sees projected effects with uncertainty from their knowledge, winners and losers by cohort with error bars, timing and political cost. Cohorts without instrument coverage show as unknown. The preview never shows truth.

Durability across elections

- Every policy has an entrenchment score.
- After a lost election, a government of a different ideology runs a term with constrained levers and may repeal weakly entrenched policies.
- Policies entrench by building a constituency: green jobs create a cohort that defends them (policy feedback, Pierson).
- The winning long-term strategy is to make climate policy hard to undo, not just to pass it.

Emergent dynamics: memory, perception and verification

Seven mechanics that make the climate lesson emerge from the simulation rather than from scripts. Three of them (collective memory, pilot policies, the levee effect) need almost no new systems, only couplings between the boxes above.

Mechanic	What happens	Where it lives	Real-world basis
Collective memory	Named events persist in the memory of cohorts who lived through them and fade as those cohorts age out	PopSim + event log	Adaptation amnesia

Mechanic	What happens	Where it lives	Real-world basis
Shifting baselines	Each generation judges nature against what it saw as children	PopSim	Shifting baseline syndrome
Pilot policies	Trial a policy in one region and measure it against similar regions	Levers + population knowledge	Natural and randomised experiments
Verification diplomacy	Treaties hold only as far as emissions can be checked	Attribution ledger + diplomacy	Measurement, reporting and verification (MRV)
Markets as sensors	Insurance and property prices signal risk before governments admit it	PopSim economy + population knowledge	Climate risk pricing
Levee effect	Protection invites development behind it, raising future losses	PopSim settlement + PlanetSim	Levee effect
Place attachment	Cohorts resist leaving sacred or ancestral places; heritage can drown	PopSim + event naming	Place attachment research

Collective memory

- Each cohort keeps a memory store of named events it experienced, weighted by severity and personal exposure.
- Memory raises perceived risk for that hazard: support for dikes, avoidance of the floodplain, acceptance of building codes.
- Memory decays as the cohort ages and is replaced; children inherit a fraction through family transmission.
- Levers slow the decay: high-water marks on buildings, memorials, anniversaries, school curricula. They are cheap but compete for attention.
- A forgotten hazard returns as a new named event, and the debrief shows the memory curve beside it.

Shifting baselines

- Each cohort stores an environmental reference point (forest cover, fish stocks, snow days, summer heat) set during its childhood.
- Perceived loss = current state compared with that reference, not with pre-industrial truth. Children born after the forests vanished do not miss them.
- Support for restoration scales with perceived loss, so each generation asks for less.
- A "grandparents' view" lens shows the state remembered by the oldest living cohort, and the comparison slider can reach back further with historical records.

Pilot policies

- Any lever with regional scope can run as a pilot in chosen regions for a set number of years.
- The population knowledge system compares pilot regions with matched control regions and returns an effect estimate with error bars.
- Good pilots narrow the uncertainty in lever previews for the national rollout. Badly matched pilots give biased estimates; social-science research improves matching.
- Pilots cost political capital in the pilot region and can create a local constituency that later defends the policy.

Verification diplomacy

- Each civ reports its emissions; the attribution ledger knows the truth.
- Other civs can check reports only through verification capacity: satellites, inspectors, shared monitoring networks, trade data.
- Under-reporting is a real strategy until caught. Exposure costs diplomatic favour and treaty standing; false accusations cost credibility.
- Shared monitoring networks are diplomatic assets that raise trust between members and lower the cost of deeper treaties.

Markets as early-warning sensors

- Insurers price hazard from observed losses and their own models; property prices respond to premiums and perceived risk.
- These signals arrive before official science admits the risk, so they work as a free leading indicator in the population knowledge system.

- Threshold events: "the coast becomes uninsurable", a property crash in a lowland region, a public reinsurance bailout.
- The government can suppress signals (price caps, public insurance of last resort), which hides risk and builds a fiscal liability.

Levee effect

- PopSim settlement and investment respond to perceived risk, not true risk.
- Building a dike lowers perceived risk, so population and capital move in behind it.
- When an event exceeds the design standard, losses are far larger than without the dike.
- Nothing is scripted: the effect emerges if settlement uses perceived risk. The debrief traces the causal chain from dike to development to disaster.

Place attachment and heritage

- Cohorts carry attachment to places: ancestral land, old towns, sacred sites. Attachment lowers the propensity to exit even when staying is costly.
- Managed retreat therefore costs more political capital where attachment is high, and is resisted as a fairness issue.
- Heritage sites have identity value for many cohorts. Their loss to the sea is a named cinematic event that lowers wellbeing beyond the material damage.
- Levers: relocate or document heritage, protect it at high cost, or let it go with a ceremony that softens the trust loss.

Procedural cities and society assets

No city is authored: every settlement is generated from what its society has discovered, adopted, can afford, has nearby and has lived through. The look of a place is therefore a readable record of its history, and two cities on different continents never look alike.

Inputs

Five kinds of state decide what a place looks like; the style pack decides only how it is drawn.

Input	Source	Shapes
Discovered	research (technology state per civ)	which building parts exist at all: no steel tower before steel and lifts
Adopted	diffusion per region and cohort (Rogers/Bass)	what share of new buildings use a part: electric light spreads from rich districts outward
Affordable	income bands per cohort, regional wealth	density, finish, informal housing, upkeep
Local	PlanetSim: geology, biome, climate, hazards	materials (timber, stone, brick, adobe), roof shape, shading, raised floors
Lived through	history store: damage, migration, named events	rebuilt districts, ruins, memorials, high-water marks, abandoned streets

The player never places these buildings. Investments and zoning (ADR-0022) steer where growth goes; the society decides what gets built.

Generation pipeline

A city is built top-down in six steps; the first four decide *what* exists from simulation state, and only the last two ask the style pack how it looks.



city generation · 6 steps, region to props

The split matters for modding: a cartoon pack and a realistic pack produce the same districts, streets and build dates, and differ only in the parts they draw.

History by accretion

Cities grow by adding blocks, not by replacing themselves, so old cores keep their old buildings and every ring of growth shows the technology of its decade.

- **Each block has a rank** in its cell, drawn once from the presentation seed `(world_seed, cell_id, block)`. Rank orders blocks from the best site (flat, near water and transport) outward.
- **A block's build year** is the first year the cell's urban fraction exceeded the block's rank. That comes from the snapshot history, so it is deterministic and needs no new simulation state.
- **Its parts** are the region's discovered set and adoption shares *in that year*, not today's.
- **Rebuilding** happens only when the state says so: a damage event destroys it, or density demand rises past what its era allows (redevelopment). The new block takes the parts of its rebuild year.
- **Decline** runs the same rule backwards: when the urban fraction falls, the highest-ranked blocks are abandoned first, and the edge of town empties before the core.

The result: a 1950 city shows a brick core from the 1890s, a tram-era ring, and concrete suburbs, and a flooded district rebuilt in 1971 stands out from its neighbours. Nobody authored any of it.

Research follows the planet, not Earth's history

Research discovers *capabilities*, and each capability can be reached by several *routes*. Which route a society finds first depends on what its land, sea and climate offer and on what its people already do, so its history can leave ours: a society on warm, nutrient-rich shallow seas can build its industry on algae and never become a coal society.

- **Capabilities, not inventions.** Turn heat into work, store and move energy, make structural material, transmit power, fix nitrogen, compute. The capability graph is shared by every society; the routes to each capability are not.
- **Routes are found by exposure.** A route's discovery chance rises with contact with its resource: fishing and seaweed farming lead to algae cultivation, mining leads to coal and metals, volcanic land leads to geothermal. This is the Eurekas idea (research follows what the society does), grounded in PlanetSim.
- **The planet decides what is possible.** Algae need sunlight, warm water, nutrients and a shallow shelf; coal needs ancient swamp basins in the geology; wind and hydro need the simulated wind and runoff. A route never appears where the planet cannot support it.
- **Lock-in makes paths stick.** Infrastructure lasts decades, towns and jobs grow around it, and research effort follows the installed base. Once a society runs on algae, coal may stay undiscovered, or be found and never pay.
- **No clean route.** Every route passes through the same physical fluxes as any other activity (no "+3 climate damage"). Algae fuel is close to carbon-neutral, but its nutrient run-off can make coastal dead zones and its ponds take shoreline and wetlands. Coal is dense and cheap but warms the planet. Different paths lead to different crises, never to a winning one.
- **Routes are data.** Each route is a record: required capabilities, resource and planet conditions, discovery triggers, physical inputs and outputs, and the set of building parts it unlocks. Earth's history is one dataset among others, and mods can add routes.

Capability	Coal path (our history)	Algae path	Other planet-driven routes
Turn heat into work	coal steam: brick mills, tall chimneys, coal yards	algae oil and biogas engines: fermentation tanks, low vents, no soot	geothermal steam in volcanic zones: insulated pipes, steam vents
Store and move energy	coal trains, gasworks	tanker barges on canals, biogas domes	pumped water in steep valleys
Structural material	brick, then steel and concrete	timber and lime early; algae-derived binders and fibre panels later	volcanic stone; adobe in dry lands
Light and power	gas lamps, then an electric grid fed by coal	biogas lamps, then a grid fed by algae plants	hydro dams where runoff allows
Food and fertiliser	mined phosphate, synthetic nitrogen	algae feed and fertiliser, coastal raceway farms	river-silt farming on floodplains
Where cities grow	inland, on coal seams and railways	ribbon cities along warm coasts and lagoons, green canals	along rivers with water power
Signature on the planet	soot, haze, warming	coastal blooms, dead zones, lost wetlands	dams, drowned valleys, changed rivers

A route that is discovered but not adopted shows in a few showcase buildings; one adopted widely becomes the ordinary look of a decade. Industrial Dawn (about 1880) starts with coal steam already known, so its paths diverge forward from there; a pre-industrial or procedural scenario lets them split from the start. The readability harness gains a signal: a coal city and an algae city must be told apart in every style.

Every route has a counterweight

No route is free: each one takes space, energy and living room from something else, and the bill arrives through the simulation, not through a penalty number. A route record must declare its counterweights in physical terms before it can ship.

Worked example: the algae society

Counterweight	What happens	Where the simulation carries it
Space	algae farms need large areas of sunny, shallow, warm water, because photosynthesis turns only a small share of sunlight into fuel	shelf area per unit of output; ponds and pens drawn on the coast
Borrowed sea	every farm takes shelf from fish, seagrass, reefs and the animals that live there	fish stocks in the food system fall; biodiversity index drops; fisher cohorts lose income and protest
Costly processing	harvesting, drying and extracting oil take much energy, so the net gain is small and a share of output feeds the process itself	energy return per route; refineries and their own power demand
Chemicals	extraction and refining use solvents and leave chemical waste	pollution stocks for water and soil; toxic run-off near refineries
Rotting algae	old or escaped algae wash up and decay, releasing foul and toxic gases such as hydrogen sulphide and methane	air quality and health exposure on the coast; methane as a greenhouse gas in the carbon budget
Nutrients and dead zones	fertilised farms and blooms strip the water of oxygen when they die	`ocean_tint` turns hypoxic; fish kills; beaches closed

An algae society therefore trades a warming atmosphere for a dying coast, foul air in its harbour towns and angry fishing communities. That is a different crisis, not an escape.

Guardrails for all routes

- **The fossil trap is always present.** The planet generator always lays down a cheap, dense, dirty energy store (coal or oil in old sedimentary basins). A society may delay it or find another way, but it always has to refuse it, so the climate lesson holds on every planet.

- **Physics limits every route.** Each record carries real yield, energy density, energy return and area per unit of output; no route may beat conservation of energy or known efficiency limits.
- **Function stays readable.** One visual language per function (stacks and domes for energy, small repeated shapes for homes, long sheds for industry); a route changes only materials, colour and detail.
- **Waste is drawn.** Slag, ash, rotting biomass, chemical ponds, landfills, and later recycling districts, each from its route's waste stream.
- **Routes travel and can be lost.** Trade, migration and espionage spread routes between civilisations and create hybrids; collapse or a broken school system can make a society forget one, leaving machines nobody can run.
- **Names and history.** Technologies and cities get generated names from each identity group's language, reusing the event-naming grammar; the debrief tells the society's own history of science ("the algae age, 1910–1990").
- **Balance is tested, not hoped for.** An automated sweep over many planet seeds checks that every route is viable somewhere, none dominates everywhere, and the fossil trap stays tempting.
- **Cities are cached.** Generated cities are cached by seed and history; at planet scale a city is only colour and lights, and full buildings appear at region and local scale.

More routes and how they turn on their society

Routes should hurt their own societies in delayed, self-inflicted ways, because that is what makes climate change hard in real life. The strongest mechanics for the game's purpose are the climate sensitivity of routes, hidden side effects and route identity.

Route catalogue

Route	Found where	Counterweight
Algae	warm, sunny, nutrient-rich shallow seas	see the worked example above
Geothermal	volcanic land	small induced earthquakes, sulphur gases, ground sinking, fields that cool down over decades

Route	Found where	Counterweight
River dams	rainy mountains	blocked fish migration, silt trapped so the delta sinks, drowned villages, methane from reservoirs
Wood and charcoal	forests	deforestation, eroded hills, smoky towns, timber shortage pushing the society elsewhere
Animal oil (whales or an invented sea creature)	rich oceans	population collapse and extinctions, counted as permanent losses
Peat	cold wetlands	fast carbon release, bog fires, subsiding land
Coal and oil (the fossil trap)	old sedimentary basins	dense, cheap, warms the planet; always present

Mechanics

- **The climate a route causes can break the route.** Each route has a climate sensitivity: algae grow better as the sea warms, until toxic blooms take over; dams lose power in droughts; thermal plants run short of cooling water in heat waves. A society can undermine its own energy source.
- **Hidden side effects discovered late.** Some routes carry dangers nobody knows at first, like lead, asbestos or ozone-destroying gases in our history. Only science reveals them, decades later; then come bans, clean-up costs and land poisoned for generations. This uses the science-as-unlock mechanic.
- **Identity grows around the route.** Mining towns, algae coasts and dam-builders' valleys develop pride, songs and political weight. That is what makes changing route hard: the fishers who lost the sea to algae and the algae workers defending their jobs both resist. It feeds PopSim identity ties and trust.
- **Depletion makes routes harder over time.** The best deposits go first; mines deepen, algae bays crowd, and each unit of energy costs more to get. The map shows deeper pits and ghost towns, and transitions happen without scripting.
- **Efficiency raises consumption (rebound).** A cheaper, cleaner route makes energy cheap, so people use more: bigger houses, more travel, more goods. Cities grow visibly more comfortable and part of the environmental gain disappears.

- **Trade dependency between civilisations.** An algae society exporting oil to a coal society creates dependency, price shocks and leverage, and can export its pollution by farming a poorer neighbour's coast.
- **An invented biosphere per planet (later).** Each planet generates a few organisms with their own traits (growth rate, preferred climate, toxins, uses), so each planet's routes are truly its own. The most ambitious idea; it can wait until the route system works with fixed organisms.

Society on the street

The city also shows who lives in it and how they feel, drawn from PopSim cohorts rather than decoration.

- **Inequality:** the share of each income band sets the mix of districts. A wide bottom band grows informal settlements on the city edge and on floodplains, where land is cheap and risk is high; that exposure is real in the simulation, not a picture.
- **Identity:** each invented identity group (fictional, never a caricature of a real community) gets its own ornament set: roof colours, decorative motifs, places of worship or assembly. Mixed regions show mixed streets.
- **Unrest:** protests, strikes and riots appear as crowds, barricades and closed shops in the districts of the cohorts involved, scaled by the event's size.
- **Migration:** arrivals show as tents and new informal blocks at the edge; departures leave shuttered houses and overgrown lots.
- **Memory:** named disasters leave marks that fade with the generation that saw them: memorials, high-water lines on walls, plaques. Their slow disappearance is the collective-memory mechanic made visible.
- **Prosperity:** upkeep follows regional wealth: fresh paint and trees in good decades, peeling facades and broken lights in bad ones.

Tests, data and build order

Truthfulness tests (design §42.1: detail never contradicts the state)

- A sampled city never shows a part whose technology its region has not discovered by the block's build year.
- The share of blocks using an adopted part matches the adoption share within a tolerance.

- Built-up area per cell matches `urban` within a tolerance; abandoned blocks match the fall in urban fraction.
- Same seed and same history give the same city on every machine.

Data needed from the simulation

- Per region: discovered technology set with discovery dates, adoption share per technology, income-band shares, identity-group shares. These belong in ADR-0012 (society state and regions); technology is a P2 subsystem.
- Per cell: `urban`, `farmland`, `mining` history from the snapshot store, plus the PlanetSim channels already planned.
- Events: protests, migration and damage, already listed in §31.2 of the specification.

Build order

1. A parts kit for Industrial Dawn in the stylised style: walls, roofs and floors in timber, stone and brick; chimneys, lamps, rail.
2. The generator running on synthetic histories (a scripted urban fraction and technology timeline), with the truthfulness tests, before any society code exists.
3. Connect it to PopSim and technology state when they land, region by region.

Open questions and next steps

Open questions

- What snapshot cadence for the slow loop versus the fast loop, and how many PopSim cohorts per civ fit the performance budget?
- Does the opposition government after a lost election run as AI, or does the player keep partial control?
- How much of the social instrument set is available in the 1880 scenario, and which instruments are research unlocks?
- Does the era score show its two axes to the player, or only the outcome label?

Next steps

- [ ] Interface spec: `Snapshot`, `FieldDesc`, `ObservedField`, `EventRecord`, ledger types, as C++ headers
- [ ] PopSim update equations with starting parameters, as a milestone for Codex
- [ ] Social observation operator: belief-state filter and first instruments (census, reports, press, elections)

- [ ] Lever cost and delay table; veto-player support computed from cohorts
- [ ] Data schemas for lens definitions, event-name templates and timeline tracks